VOLUNTARY PRODUCT ACCESSIBILITY TEMPLATE · WCAG 2.1 A & AA

Accessibility Conformance Report

Modulus · Report date 2026-10

Introduction

Modulus strives to be compliant with the Web Content Accessibility Guidelines (WCAG) 2.1 AA standard, as required by the United States Department of Education and as adopted by The Ohio State University. Modulus is the learner activity database and instructor analytics surface that sits alongside the Ximera interactive textbook platform; this report covers the Modulus web application itself, not the Ximera content it observes.

Technical conformance with WCAG does not guarantee that every user with a disability will be able to complete every task. Institutions deploying Modulus should publish their own accessibility statement, and instructors using Modulus should ensure the courses and assignments they assemble in Ximera (and surface in their LMS) carry their own accessibility statements.

Product Information

Item	Details
Name	Modulus
Type	Web-based learner activity database and instructor analytics application
Description	Records time-series interaction data and page-state snapshots from Ximera assignments and presents them to instructors; integrates with institutional LMSes via LTI 1.3.
Primary Use Case	Postsecondary education — instructors reviewing assignment activity; learners launched into Ximera through their LMS
Hosting Model	Self-hosted or institutionally hosted web application
Contact	Bart Snapp, snapp@math.ohio-state.edu
Report Date	2026-10

Evaluation Methods

The assessment employed:

- Manual expert review by the Modulus development team
- An external accessibility audit of the gradebook application, whose findings (keyboard operation of tabs and pagination, accessible names, focus-indicator contrast, heading and list semantics, page titles, and chart text alternatives) have been addressed
- Keyboard-only navigation across the public site, LTI launch flow, instructor dashboards, and administrative forms
- Targeted screen-reader testing with NVDA (Firefox, Chrome on Windows) and VoiceOver (Safari, Chrome on macOS)
- Automated linting (Biome) and component-level accessibility checks during development

- Visual inspection of color contrast in default and dark themes

The report reflects the current state of the application. Part of the audit fixes have been verified in the browser; the rest, including the dashboard navigation, pagination, and sortable-table fixes, are complete and awaiting a signed-in verification pass. The reworked charts have passed visual review and automated component tests, and their NVDA and VoiceOver verification is still in progress.

Applicable Standards

Standard	Included
WCAG 2.1 Level A	Yes
WCAG 2.1 Level AA	Yes
WCAG 2.1 Level AAA	Not Evaluated

Conformance Terminology

- **Supports**: Functionality meets the criterion without known defects.
- **Partially Supports**: Some functionality does not fully meet the criterion.
- **Does Not Support**: The majority of functionality does not meet the criterion.
- **Not Applicable**: The criterion does not apply to the product.

WCAG 2.1 Level A Success Criteria

Criterion	Conformance	Remarks
1.1.1 Non-text Content	Supports	Icons and decorative images carry appropriate alt text or are hidden from assistive technology. Each chart is exposed as a single image with a native `<title>` and `<desc>`, accompanied by a visible text summary and a disclosable data table that gives every exact value.
1.2.1–1.2.3 Audio/Video (Prerecorded)	Not Applicable	Modulus does not host audio or video content.
1.3.1 Info and Relationships	Supports	Pages use semantic HTML landmarks (`header`, `nav`, `main`, `footer`), heading levels, lists, and tables for tabular data. Form fields are programmatically associated with their labels and help text (`aria-describedby`). Sortable table columns expose `aria-sort`, and grouped content such as members and statistics uses list semantics.
1.3.2 Meaningful Sequence	Supports	Content order in the DOM matches the intended reading and tab order.
1.3.3 Sensory Characteristics	Supports	Instructions and call-outs do not rely on shape, size, location, or sound alone.
1.4.1 Use of Color	Supports	Status (e.g. submission state, error/success messages) is conveyed by text and icon in addition to color. Chart series are distinguished using a combination of color, legend labels, and tooltips; chart styling is adjusted as needed to ensure information is not conveyed by color alone.

Criterion	Conformance	Remarks
1.4.2 Audio Control	Not Applicable	No auto-playing audio or video.
2.1.1 Keyboard	Supports	All interactive controls, including the navigation drawer, dialogs, tables, and forms, are reachable and operable via the keyboard. Tabs switch with the arrow keys and every pagination control is a real button or link. Chart values are available in a native data table, so no information depends on hovering a chart.
2.1.2 No Keyboard Trap	Supports	Focus can be moved into and out of every component, including modal dialogs, using standard keyboard controls.
2.1.4 Character Key Shortcuts	Not Applicable	Modulus does not implement single-character keyboard shortcuts.
2.2.1 Timing Adjustable	Supports	No time limits are imposed on instructor tasks. Authentication sessions extend silently within their LTI launch window.
2.2.2 Pause, Stop, Hide	Not Applicable	No moving, blinking, scrolling, or auto-updating content beyond brief loading indicators.
2.3.1 Three Flashes or Below	Supports	No content flashes more than three times per second.
2.4.1 Bypass Blocks	Supports	A "Skip to main content" link is provided, and page regions are marked up as ARIA landmarks.
2.4.2 Page Titled	Supports	Each route sets a descriptive `<title>` through its route `head`, and the title updates on every client-side navigation.
2.4.3 Focus Order	Supports	Tab order follows the logical reading sequence; dialogs trap focus while open and restore it on close. After a client-side navigation, focus moves to the new page's main heading.
2.4.4 Link Purpose (In Context)	Supports	Link text is descriptive on its own or within its immediate context; ambient "click here" links are avoided.
2.5.1 Pointer Gestures	Not Applicable	All functions are operable via single-point activation; no multi-point or path-based gestures are required.
2.5.2 Pointer Cancellation	Supports	Actions are triggered on the up-event of a pointer, allowing users to abort by moving off the target.
2.5.3 Label in Name	Supports	Visible labels and the accessible name of each control match.
2.5.4 Motion Actuation	Not Applicable	Modulus does not use device motion or user motion as input.
3.1.1 Language of Page	Supports	The document `lang` attribute is set on every page (currently English).
3.2.1 On Focus	Supports	Receiving focus does not cause an unexpected change of context.
3.2.2 On Input	Supports	Changing a field value does not submit the form or navigate the user; explicit submit actions are required.
3.3.1 Error Identification	Supports	Form errors are identified in text near the offending field, in addition to color and icon cues.
3.3.2 Labels or Instructions	Supports	All form fields have visible labels; additional instructions are provided where input format is constrained.

Criterion	Conformance	Remarks
4.1.1 Parsing	Supports	The rendered HTML conforms to modern parsing requirements (the criterion is treated as obsolete in WCAG 2.2).
4.1.2 Name, Role, Value	Supports	Native form controls expose name, role, and value automatically. Custom components built on Base UI expose ARIA attributes, and icon-only controls (copy, search, clear, and close buttons), select filters, and forms carry descriptive accessible names. Charts are not interactive for assistive technology.

WCAG 2.1 Level AA Success Criteria

Criterion	Conformance	Remarks
1.2.4 Captions (Live)	Not Applicable	No live audio or video.
1.2.5 Audio Description (Prerecorded)	Not Applicable	No prerecorded audio or video.
1.3.4 Orientation	Supports	Layouts reflow in both portrait and landscape orientations; no orientation is locked.
1.3.5 Identify Input Purpose	Partially Supports	Email and password fields use the matching input types, but fields that collect user information do not yet carry `autocomplete` tokens (`email`, `name`, `current-password`), and several account forms disable autocomplete. Adding the tokens is planned.
1.4.3 Contrast (Minimum)	Supports	Body text and primary controls meet the WCAG AA 4.5:1 (normal text) and 3:1 (large text) minimums in both light and dark themes. Chart colors are tuned to meet contrast targets, and every chart value is also available as text in its data table.
1.4.4 Resize Text	Supports	Text can be resized to 200% without loss of content or functionality.
1.4.5 Images of Text	Supports	Text is rendered as text; logos are the only image-of-text exception.
1.4.10 Reflow	Supports	Content reflows at 400% zoom without requiring two-dimensional scrolling, except where two-dimensional layout is essential (data tables, charts).
1.4.11 Non-text Contrast	Supports	Focus indicators (including those on secondary buttons, cards, breadcrumbs, and the footer), form borders, and primary buttons meet the 3:1 contrast minimum against adjacent colors. Chart axes, gridlines, and series strokes are tuned to meet contrast targets and adjusted as needed.
1.4.12 Text Spacing	Supports	User stylesheets that adjust line height, paragraph spacing, letter spacing, and word spacing do not break the layout.
1.4.13 Content on Hover or Focus	Supports	Tooltips and popovers are dismissible with Escape, remain visible while hovered, and persist until the triggering element loses hover or focus. Chart pointer tooltips are supplemental; their values also appear in the chart's data table.
2.4.5 Multiple Ways	Supports	Pages are reachable via primary navigation, deep links from the LMS, and direct URLs.

Criterion	Conformance	Remarks
2.4.6 Headings and Labels	Supports	Headings and form labels describe the topic or purpose of their section or control.
2.4.7 Focus Visible	Supports	A visible focus indicator is provided for every keyboard-operable control, including tabs, search buttons, checkboxes, and breadcrumb links, and it renders consistently across browsers; the indicator meets non-text contrast minimums.
3.1.2 Language of Parts	Not Applicable	All in-product text is in a single language (English).
3.2.3 Consistent Navigation	Supports	Navigation and chrome are presented consistently across pages within a given role (public, instructor, admin).
3.2.4 Consistent Identification	Supports	Components that perform the same function (e.g. "Save", "Cancel", "Export") are identified consistently.
3.3.3 Error Suggestion	Supports	Where a form value is known to be incorrect, a suggestion is offered (e.g. expected format, allowed values).
3.3.4 Error Prevention (Legal, Financial, Data)	Supports	Destructive administrative actions require explicit confirmation; no financial or legal transactions are handled by Modulus.
4.1.3 Status Messages	Partially Supports	Toast notifications, inline form-status messages, chart loading and error states, and client-side page changes are exposed via ARIA live regions. A small number of asynchronous updates in instructor dashboards do not yet emit status messages and are being addressed.

Known Issues and Roadmap

The following areas have been identified for ongoing accessibility work:

- **Live region coverage** — Extending ARIA live-region announcements to all asynchronous status changes in instructor dashboards.
- **Input purpose** — Adding `autocomplete` tokens to the fields that collect user information.
- **Chart verification** — Completing the NVDA and VoiceOver browser checks of the reworked charts.

Legal Disclaimer

This document is provided for informational purposes only and does not constitute a warranty of accessibility compliance. Accessibility outcomes vary based on configuration, deployment, integrations, and end-user environments. The Modulus project will continue to improve accessibility as part of its development roadmap and welcomes accessibility feedback via its public issue tracker.